Recap : Linear System Solvers

Exercise sheets : the first one is out (IEEE-754 & linear systems), solutions released with the next sheet. Purpose : ungraded extra practice material.

Key idea

Don't invert (big) matrices : $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ usually **does not mean** "compute an inverse of $\mathbf{A}$". Read it as "call a linear system solver to solve $\mathbf{Ax} = \mathbf{b}$".

What kind of linear system solver ?

— **Diagonal** : `for i in range(n): x[i] = b[i] / A[i, i]`
— **Permutation** : `for i in range(n): x[i] = b[perm[i]]`
— **Triangular** : forward/backward substitution.
— **Dense** : general solver, e.g. LU factorization.

Key formulas

Recap : $\mathbf{A} = \mathbf{LU}$, $\mathbf{U} = \mathbf{E}_6\mathbf{E}_5\mathbf{E}_4\mathbf{E}_3\mathbf{E}_2\mathbf{E}_1\mathbf{A}$; for SPD matrices, Cholesky $\mathbf{A} = \mathbf{LL}^T$ (half as expensive).

Example

Solving an upper triangular system (back substitution)

```
def solve_upper(A, b):
    n = len(b)
    x = np.copy(b)
    for i in reversed(range(n)):
        x[i] /= A[i, i]
        x[:i] -= A[:i, i] * x[i]
```

Key idea

Cost ($\mathbf{A} \in \mathbb{R}^{10000\times10000}$) : **factorization** $\mathcal{O}(n^3)$, **substitution** $\mathcal{O}(n^2)$. Usage of $\mathbf{PLU} = \mathbf{A}$: solve $\mathbf{Px}_1 = \mathbf{b}$ (permute), then $\mathbf{Lx}_2 = \mathbf{x}_1$ (forward subst.), then $\mathbf{Ux}_3 = \mathbf{x}_2$ (backward subst.) – `la.lu_factor` / `la.lu_solve`, or `la.lu` + `la.solve_triangular`.

Examples of Linear Systems

Example

Ideal ballistic motion (a basketball throw) : gravity + initial velocity give a parabola

$$y = ax^2 + bx + c$$

Step 1 : think about observables and unknowns. For a measured position (x_1, y_1) : y_1 and x_1 are *known*, a, b, c are *unknown*.

Definition

REGRESSION

— We have a model of the **expected behaviour**, containing unknown parameter values.
— We have data / measurements that *may* be supported by the model.
— **Goal** : find parameters that best support the data.

If this can be expressed as the solution of a linear system : **linear regression**.

Key formulas

How many data points (measurements) do we need ? The equation is **linear in the unknowns** : 3 measurements give 3 equations,

$$\begin{bmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ x_3^2 & x_3 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

Definition

POLYNOMIAL REGRESSION (the basketball is a representative of this larger class)
Find a polynomial that *perfectly interpolates* a given set of points : $y(t_i) = p_i$.

$$\begin{bmatrix} t_0^3 & t_0^2 & t_0 & 1 \\ t_1^3 & t_1^2 & t_1 & 1 \\ t_2^3 & t_2^2 & t_2 & 1 \\ t_3^3 & t_3^2 & t_3 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix}$$

The matrix is a **Vandermonde matrix**.

Warning

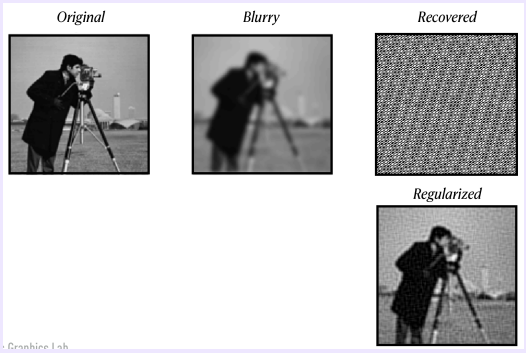
Food for thought – what could possibly go wrong ?

— What if the observations are *weak* (all very close together) ?
— What if there are > 3 observations ? Use a subset ? Which ones ? Use all somehow ?
— What if there is noise ? What if the model violates the assumptions ?

A "Bad" Linear System : Deconvolution

Example

Deconvolution : removing blur from a signal (e.g. an image) – "can you enhance that ?". Solving the linear system exactly gives pure noise ; a *regularized* solution recovers a recognizable image.



Forward and Backward Error of Solvers

Key idea

Recap : backward error in the input space, forward error in the output space.

$$\text{Condition number : ratio } \frac{\text{forward error}}{\text{backward error}}$$

Key formulas

Applied to linear system solving

- Theoretical infinite precision solver : $\mathbf{x} = f(\mathbf{b}) = \mathbf{A}^{-1}\mathbf{b}$. Its inverse is simple : $f^{-1}(\mathbf{x}) = \mathbf{A}\mathbf{x}$.
- Actual finite precision solver : $\hat{\mathbf{x}} = \hat{f}(\mathbf{b})$ – how accurate is $\hat{\mathbf{x}}$? Natural to expect $f^{-1}(\hat{\mathbf{x}}) = \mathbf{A}\hat{\mathbf{x}} \approx \mathbf{b}$.

Definition

RESIDUAL

$$r(\hat{\mathbf{x}}) = \mathbf{A}\hat{\mathbf{x}} - \mathbf{b}$$

Gives an approximation of the **backward error** of a linear system solver.

The Matrix Norm

Interlude : a matrix maps the points of a circle to an ellipse (an ellipsoid in higher dimensions).

Definition

MATRIX NORM (aka. operator norm)

$$\|\mathbf{A}\| = \max\{\|\mathbf{A}\mathbf{x}\|, \text{ where } \|\mathbf{x}\| = 1\}$$

Determines the maximum radius of the ellipse that $\mathbf{A}$ makes of the unit circle.

Key formulas

Some inequalities

$$\|\mathbf{A}\mathbf{x}\| \leq \|\mathbf{A}\|\,\|\mathbf{x}\|$$

The length of a transformed vector is bounded by its original length and the most that it can be stretched by $\mathbf{A}$.

$$\|\mathbf{A}\mathbf{B}\| \leq \|\mathbf{A}\|\,\|\mathbf{B}\|$$

Successive application of $\mathbf{A}$ and $\mathbf{B}$ cannot stretch a vector by more than the product of their individual bounds.

Condition Number of a Linear System

Key formulas

Solving a perturbed linear system

- Original system : $\mathbf{A}\mathbf{x} = \mathbf{b}$
- Perturbed system : $\mathbf{A}(\mathbf{x} + \Delta\mathbf{x}) = \mathbf{b} + \Delta\mathbf{b}$
- What we'd like to know : how big is $\|\Delta\mathbf{x}\|/\|\mathbf{x}\|$ (relative error of the solution)?

Ingredient 1 : $\|\mathbf{b}\| = \|\mathbf{A}\mathbf{x}\| \leq \|\mathbf{A}\| \cdot \|\mathbf{x}\| \Leftrightarrow \|\mathbf{x}\| \geq \frac{\|\mathbf{b}\|}{\|\mathbf{A}\|}$
Ingredient 2 : $\|\Delta\mathbf{x}\| = \|\mathbf{A}^{-1}\Delta\mathbf{b}\| \leq \|\mathbf{A}^{-1}\| \cdot \|\Delta\mathbf{b}\|$
Putting it together :

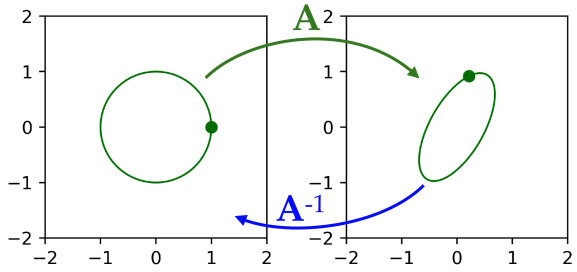
$$\frac{\|\Delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq \underbrace{\|\mathbf{A}^{-1}\| \cdot \|\mathbf{A}\|}_{=\text{cond}(\mathbf{A})=\kappa(\mathbf{A})} \cdot \frac{\|\Delta\mathbf{b}\|}{\|\mathbf{b}\|}$$

Definition

CONDITION NUMBER

$$\text{cond}(\mathbf{A}) = \|\mathbf{A}\| \cdot \|\mathbf{A}^{-1}\|$$

Measures how much $\mathbf{A}$ (and $\mathbf{A}^{-1}$) distort the unit circle : the more elongated the ellipse, the larger the condition number.



Properties of the Condition Number

Summary

- $\text{cond}(\mathbf{A}) = \infty$ iff $\mathbf{A}$ is singular
- $\text{cond}(\mathbf{A}) = 1$ if $\mathbf{A}$ is the identity matrix
- $\text{cond}(\gamma\mathbf{A}) = \text{cond}(\mathbf{A})$, $\gamma \in \mathbb{R}$
- $\text{cond}(\mathbf{Q}\mathbf{A}) = \text{cond}(\mathbf{A})$ where $\mathbf{Q}$ is a rotation matrix

Warning

What about the determinant? It is *not* a good way to determine whether a linear system is easy to solve. We know that $\det(\mathbf{A}) = 0$ iff $\mathbf{A}$ is singular. But with $\mathbf{A} = \frac{1}{10}\mathbf{I} \in \mathbb{R}^{1000\times1000}$, $\det(\mathbf{A}) = 10^{-1000}$, yet this linear system is *trivial* to solve. On the other hand, $\text{cond}(\mathbf{A}) = 1$ (!)

Key idea

Implications (stated without proof) : if you are using a good linear system solver, and if $\mathbf{b}$ can be assumed to be accurate, then

$$\frac{\|\Delta\mathbf{x}\|}{\|\mathbf{x}\|} \lesssim \text{cond}(\mathbf{A})\ \text{ulp}(1)$$

where $\text{ulp}(1) \approx 2^{-23}$ (single precision), $\approx 2^{-52}$ (double precision).

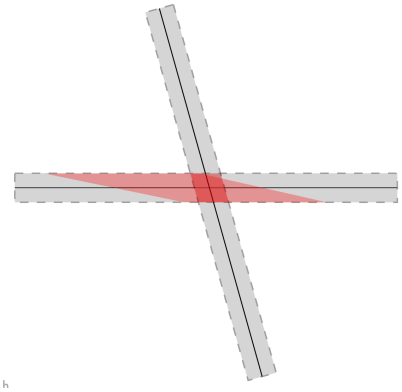
Warning

A chicken-and-egg problem : inverting matrices is expensive; we want to know the condition number to find out if this is even possible… but our definition of $\text{cond}(\mathbf{A})$ requires inverting a matrix and computing a matrix norm (how?!). Stay tuned!

Badly Conditioned Systems

Key idea

Geometry (row-based interpretation) : noise & errors may be greatly magnified. When the planes (lines) $\mathbf{a}^{(i)} \cdot \mathbf{x} = b_i$ are nearly parallel, a small uncertainty on each one gives a huge region of possible intersections.



Example

Conditioning of the Vandermonde matrix (maps polynomial coefficients to polynomial evaluations) :

$$\mathbf{A} = \begin{pmatrix} \alpha_1^n & \dots & \alpha_1^2 & \alpha_1 & 1 \\ \vdots & & \vdots & \vdots & \vdots \\ \alpha_m^n & \dots & \alpha_m^2 & \alpha_m & 1 \end{pmatrix}$$

Condition number for $n = 100$ (uniformly spaced positions on $[0, 1]$) :

$$\text{cond}(\mathbf{A}) = 7.31 \cdot 10^{19} \text{ (!!!!!!)}$$

Linear Least Squares

Definition

FROM LINEAR SYSTEMS TO LEAST SQUARES
n constraint equations (many), m parameters (few) : instead of $\mathbf{Ax} = \mathbf{b}$, ask for $\|\mathbf{Ax} - \mathbf{b}\| \approx 0$. For almost all values of $\mathbf{b}$, this *tall* system is not solvable (generally ill-defined) : try to get as close as possible.

Key formulas

Compromise : minimize the residual $\|\mathbf{r}(\mathbf{x})\| = \|\mathbf{Ax} - \mathbf{b}\|$.
1. Expand (the squared length of a vector is the inner product with itself) :

$$\begin{aligned} \|\mathbf{Ax} - \mathbf{b}\|^2 &= \langle \mathbf{Ax} - \mathbf{b}, \mathbf{Ax} - \mathbf{b} \rangle \\ &= \langle \mathbf{Ax}, \mathbf{Ax} \rangle - \langle \mathbf{Ax}, \mathbf{b} \rangle - \langle \mathbf{b}, \mathbf{Ax} \rangle + \langle \mathbf{b}, \mathbf{b} \rangle \\ &= \mathbf{x}^T \mathbf{A}^T \mathbf{Ax} - 2\mathbf{b}^T \mathbf{Ax} + \|\mathbf{b}\|^2 \end{aligned}$$

2. Take the derivative ($\|\mathbf{b}\|^2$ is constant) :

$$\frac{\partial}{\partial \mathbf{x}} \left[\mathbf{x}^T \mathbf{A}^T \mathbf{Ax} - 2\mathbf{b}^T \mathbf{Ax} + \|\mathbf{b}\|^2 \right] = 2\mathbf{A}^T \mathbf{Ax} - 2\mathbf{A}^T \mathbf{b}$$

3. Set it to zero (Carl Friedrich Gauss) :

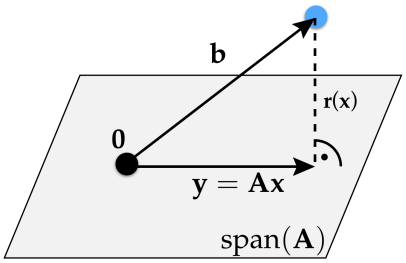
$$\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b} \qquad \textbf{(normal equations)}$$

Key idea

The "best" solution (in the L_2 sense) is given by the normal equations, which are equivalent to a **smaller square linear system** ($\mathbf{A}^T\mathbf{A}$ is $m \times m$).

Definition

WHY "NORMAL" EQUATIONS ?
$\mathbf{A}^T(\mathbf{Ax} - \mathbf{b}) = \mathbf{0}$: at the optimum, the residual $\mathbf{r}(\mathbf{x})$ is **normal** (perpendicular) to $\text{span}(\mathbf{A})$; $\mathbf{y} = \mathbf{Ax}$ is the orthogonal projection of $\mathbf{b}$.



Weekly Summary

Summary

- Pick the solver for the matrix type ; factorization $\mathcal{O}(n^3)$, substitution $\mathcal{O}(n^2)$.
- Regression : data + model linear in the unknowns $\Rightarrow$ linear system (Vandermonde matrix for polynomials).
- Residual $\mathbf{A}\hat{\mathbf{x}} - \mathbf{b} \approx$ backward error.
- Matrix norm, $\text{cond}(\mathbf{A}) = \|\mathbf{A}\| \, \|\mathbf{A}^{-1}\|$, relative error $\lesssim \text{cond}(\mathbf{A}) \, \text{ulp}(1)$; the determinant says nothing about difficulty.
- Least squares : minimize $\|\mathbf{Ax} - \mathbf{b}\|^2 \Rightarrow$ normal equations $\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}$.